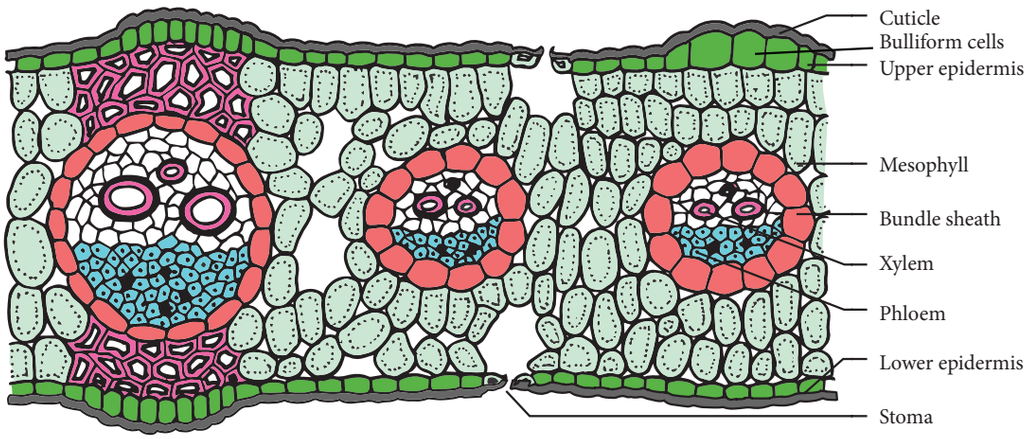



**Figure 12.7** Transverse section of Monocot Leaf

Each vascular bundle is surrounded by parenchymatous bundle sheath. Vascular bundles are conjoint, collateral and closed. Xylem is present towards upper epidermis and phloem towards lower epidermis.

**Table 12.4** Differences between of Dicot and Monocot Leaf

| S. No. | Dicot Leaf | Monocot Leaf |
|---|---|---|
| 1 | Dorsiventral leaf | Isobilateral leaf |
| 2 | Mesophyll is differentiated into palisade and spongy parenchyma | Mesophyll is not differentiated into palisade and spongy parenchyma |

## 12.9 Plant Physiology

### 12.9.1 Plastids

Plastids are double membrane bound organelles found in plants and some algae. They are responsible for preparation and storage of food. There are three types of plastids.

Chloroplast - green coloured plastids
Chromoplast - yellow, red, orange coloured plastids
Leucoplast - colourless plastids

### 12.9.2 Structure of Chloroplast

Chloroplasts are green plastids containing green pigment called **chlorophyll**. Chloroplasts are oval shaped organelles having a diameter of 2-10 micrometer and a thickness of 1-2 micrometer.

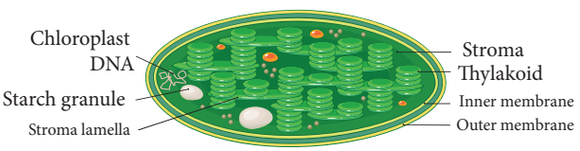



**Figure 12.8** Ultrastructure of Chloroplast

1. **Envelope:** Chloroplast envelope has outer and inner membranes which is seperated by intermembrane space.
2. **Stroma:** Matrix present inside to the membrane is called stroma. It contains DNA, 70 S ribosomes and other molecules required for protein synthesis.
3. **Thylakoids:** It consists of thylakoid membrane that encloses thylakoid lumen. Photosynthetic pigments are present in thylakoids. Thylakoids forms a stack of **disc like structures** called a grana (singular-granum).
4. **Grana:** Thylakoids arranged in the form of discs stacked one above the other called granum. Grana are inter connected by stroma lamella.



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